

Basil Kanaan

Software Engineer | Brampton, ON (Greater Toronto Area) | Open to remote

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SUMMARY

Full-stack software engineer who takes systems from an empty repository to production. Founding engineer and acting CTO at an early-stage AI startup through public launch; since then, solo-built a multi-tenant AI receptionist SaaS — event-driven Python across 13 AWS Lambda services, LLM orchestration with multi-model failover, Stripe billing, Cognito auth with RBAC, and a Nuxt 3 / TypeScript portal backed by roughly 530 automated tests. Now runs Qurra, a studio building premium hand-coded websites, and trains LLMs freelance on Python and English tasks. Works daily with Claude, Codex, and Gemini. Strongest on small teams where the problem is ambiguous and the whole stack is fair game.

TECHNICAL SKILLS

Languages: Python (3.12), TypeScript, JavaScript (ES6+), SQL, HTML/CSS

Frontend: Vue 3, Nuxt 3, Pinia, Vite, React.js, Astro, Tailwind CSS, Zod, GSAP / ScrollTrigger, interactive 3D, responsive and mobile-first design, Figma

Cloud & Backend: AWS Lambda, API Gateway, DynamoDB (+ Streams), S3, RDS, Cognito, Firebase (Auth, Firestore, Storage), EventBridge, Step Functions, CloudWatch, SAM / CloudFormation (infrastructure as code), boto3, WebSockets, REST API design, event-driven and serverless architecture, Docker, Vercel, CI/CD (GitHub Actions)

AI & LLM engineering: Google Gemini, OpenAI, Anthropic Claude, OpenRouter; function/tool calling, structured JSON output schemas, multi-agent routing and delegation, multi-model failover, conversation memory and rolling summarisation, prompt engineering and system design, golden-trajectory authoring for model training, usage metering

AI-assisted development: Claude and Claude Code, OpenAI Codex, Google Gemini, Google Flow (Veo) for generative video — used daily across architecture, implementation, code review, and asset production

Integrations & Payments: Stripe (Checkout, Customer Portal, signed webhooks), Twilio (SMS/WhatsApp), ElevenLabs voice agents, Make.com, Meta Ads, OAuth 2.0 and webhook patterns, Google Workspace, Microsoft 365, HubSpot, Asana, Slack, Notion, Dropbox, Meta (Facebook/Instagram/WhatsApp), X/Twitter

Testing & practice: Pytest, Vitest, idempotency and retry design, webhook signature verification, RBAC and audit logging, structured logging and tracing, technical specs, code review, release management, Git

EXPERIENCE

Qurra — Founder & Full-Stack Software Engineer | Aug 2025 – Present | Brampton, ON

Independent web studio — premium, hand-coded websites for local businesses (getqurra.com)

- **End-to-end delivery:** built 15+ hand-coded websites, including earlier paid freelance work for an event MC (mczee.ca) and a UK automotive shop, plus pitch builds for 7 local businesses across home building, real estate, food service, grooming, tree care, and automotive — each taken from Figma design through deployment, with no templates or page builders.
- **Motion & 3D:** build cinematic scroll-driven interfaces with GSAP and ScrollTrigger, video-backed heroes, and interactive 3D, including a true-3D concept site (aonagi.getqurra.com) that runs the same scene across phone, laptop, and desktop.
- **Infrastructure & operations:** ship every build as its own Git repository and Vercel project — client sites on their own domains, pitch builds on studio subdomains, studio properties joined under one domain via rewrite rules; manage DNS, environment configuration, metadata and Open Graph tags, analytics, and uptime.
- **AI-assisted workflow:** work daily in Claude and Claude Code, OpenAI Codex, and Gemini across architecture, implementation, and review; produce cinematic hero footage in Google Flow (Veo), including multi-shot sequences stitched from separate generations to hold continuity.
- **Client acquisition:** run sales solo — hundreds of cold calls, cold emails, and DMs, Meta paid ad campaigns tested across multiple verticals, and an Instagram channel of build reels — alongside scoping, pricing, proposals, and post-launch support.

Earlier product phase — multi-tenant AI receptionist SaaS (Aug 2025 – May 2026), designed and built solo:

- **Product & serverless backend:** AI receptionist platform handling conversations across WhatsApp, SMS, voice, and an embeddable web chat widget — automated FAQ resolution, lead capture, booking workflows, and human takeover; event-driven Python across 13 independently deployed AWS Lambda services behind API Gateway, with DynamoDB and S3 for state, provisioned through AWS SAM / CloudFormation infrastructure as code.
- **LLM orchestration:** Gemini orchestration engine with structured JSON schemas, function calling, per-tenant prompts and tools loaded from S3, rolling conversation summaries, exponential backoff, duplicate-response suppression, and automatic multi-model failover.

- **Reliability:** DynamoDB outbox with a stream-driven dispatcher for asynchronous tool execution — idempotency controls, retry scheduling, dead-letter states, failure tracking, and idempotent result callbacks into Make.com webhooks.
- **Integrations & billing:** per-tenant Twilio numbers and ElevenLabs voice agents with transcript ingestion and unified cross-channel history; Stripe Checkout and Customer Portal billing with signed webhooks, plan entitlements, usage tracking, and payment-triggered workspace provisioning.
- **Frontend, access control & testing:** Nuxt 3 / Vue 3 / TypeScript portal with Cognito authentication, omnichannel inbox, billing controls, and an RBAC-protected admin console with impersonation auditing; roughly 530 Pytest and Vitest tests across authentication, billing, idempotency, LLM failover, and API contracts; webhook signature verification, CORS and widget-origin allowlists, least-privilege deploy roles, and S3 Object Lock.

Outlier & DataAnnotation — AI Trainer (Freelance) | May 2026 – Present | Remote

Contract LLM training work, part-time alongside Qurra

- **Golden trajectories:** identify and construct golden trajectories — the ideal step-by-step path a model should follow to solve a task — for Python and English-language problems, used as reference data to train and evaluate large language models.
- **Prompt engineering:** write and refine prompts and task instructions for model training and evaluation work.

UnlockAI — Founding Engineer / Acting CTO | Oct 2023 – Aug 2025 | Remote (Dubai, UAE)

Early-stage startup building a multi-agent AI chat assistant (“OneChat”) and one-click workflow automations

- **Platform architecture (AWS + Python):** designed ChatManager and IntegrationsManager on Lambda and API Gateway with Step Functions and EventBridge; DynamoDB and RDS for state; S3 presigned uploads; structured logging; idempotent, retry-safe execution.
- **Multi-agent orchestration:** shipped “Aura,” an AI team manager that delegates to specialist agents — sequential tool calling, confirmation gates, audit logging, model routing across OpenAI, Gemini, and OpenRouter, memory via a brand ledger, WebSocket streaming, and attachment handling.
- **Integrations & automations:** documented and tested 80+ actions across Google Workspace, Microsoft 365, Meta, HubSpot, Asana, Slack, Notion, Dropbox, and X/Twitter using OAuth 2.0 and webhooks; built one-click presets such as Property Inquiry Follow-up (HubSpot lead → personalized Gmail reply → Slack/WhatsApp reminder).
- **Frontend (Vue.js):** rebuilt the OneChat and Admin interfaces for responsiveness and speed; improved streaming and message formatting, file uploads, integration run feedback, and connection management.
- **Delivery & leadership:** scoped features from client, mentor, and investor discovery calls; led a core team plus five contractors; authored specs, ran code reviews, and drove releases, including the 2025 pivot from an earlier social-media scheduler to automations.
- **Outcomes:** public launch in April 2025; 235 sign-ups and roughly 100 monthly active users by July 2025; prepared seed materials with activation and usage metrics.

Career Break — Family Caregiving | Jul 2022 – Dec 2022

Earlier: Programming Instructor, Code Ninjas (Jun – Jul 2020); freelance Java plugin developer for Minecraft servers (Jul – Aug 2019).

INDEPENDENT PROJECTS

- **Areté OS — areteos.vercel.app (2026, solo, live with early users):** self-improvement app where Gemini turns a user's goals into a blueprint of traits and measurable standards, then scores each journal reflection against it. React 19, TypeScript, Firebase, and Vercel serverless functions with model fallback and per-plan rate limits; Stripe subscriptions with signed, idempotent webhooks; ElevenLabs speech-to-text voice commands; Firestore security rules.
- **LoLHighlights (Python, OpenCV; 2023–2024):** scans gameplay video and timestamps highlights by template-matching on-screen multi-kill banners across sampled frames; sped up extraction by moving to decord and classifying frames in parallel with a thread pool.
- **2D-BattleRoyale (JavaScript; 2023):** browser game with AI opponents, live on GitHub Pages; refactored and retuned from a multiplayer version co-built with a teammate at the University of Toronto (2021–2022; React, Express, WebSockets, PostgreSQL).

EDUCATION

Honours Bachelor of Science, Computer Science Specialist

University of Toronto — Graduated with High Distinction | CGPA 3.52 / 4.0 | June 2022